

BIODIVERSITY

Objectives

Classification

- identify organisms using diagnostic features of the five kingdoms
- use diagnostic features to divide kingdoms into phyla
- state the taxonomic hierarchy
- observe the rules of binomial nomenclature

Importance of biodiversity

- describe socio economic importance of the five kingdoms

Socio-economic importance of

1. Kingdom Prokaryote

- fermentation
- biotechnology
- food spoilage
- decomposition

2. Kingdom Protista

- Plasmodium sp - malaria
- Schistosoma sp - schistosomiasis
- Trypanosoma sp - Trypanomiasis

3. Kingdom Fungi

- fermentation
- penicillin production
- decomposition
- food spoilage
- food

4. Kingdom Plantae

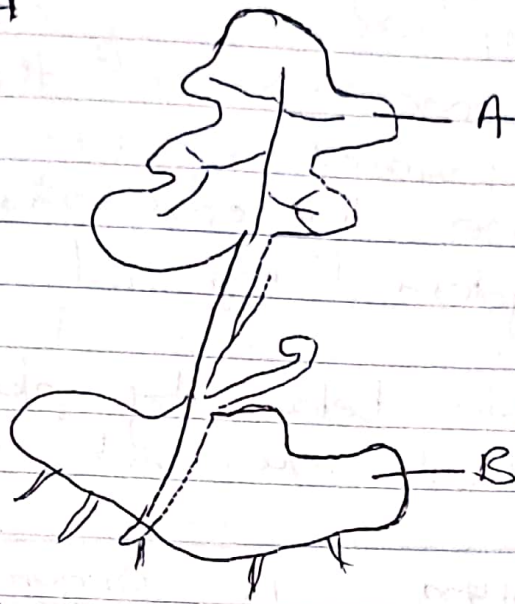
- producers

- carbon sink
- timber
- medicinal use
- tourism

S. Kingdom Animalia

- tourism
- food
- hunting
- leather
- fishing

2 9a. Fig below shows alternation of generation in a plant



- ai Identify the phylum to which the organism belongs [1]
- ii Name the parts labelled A and B [2]
- iii Suggest any 3 features that enable the organism in this phylum to survive on land [3]

2/2 10a. Fig below shows the larval stage of a butterfly.

- i State the phylum to which this organism belongs
- ii List any 2 characteristic features of this phylum [2]
- iii Explain the importance of the organism in [3]

20/2 10a. State the rules of binomial nomenclature [2]

- b. Explain any 2 important roles of the kingdom Plantae [2]

10/2 10. Fig below shows an organism in the kingdom animalia

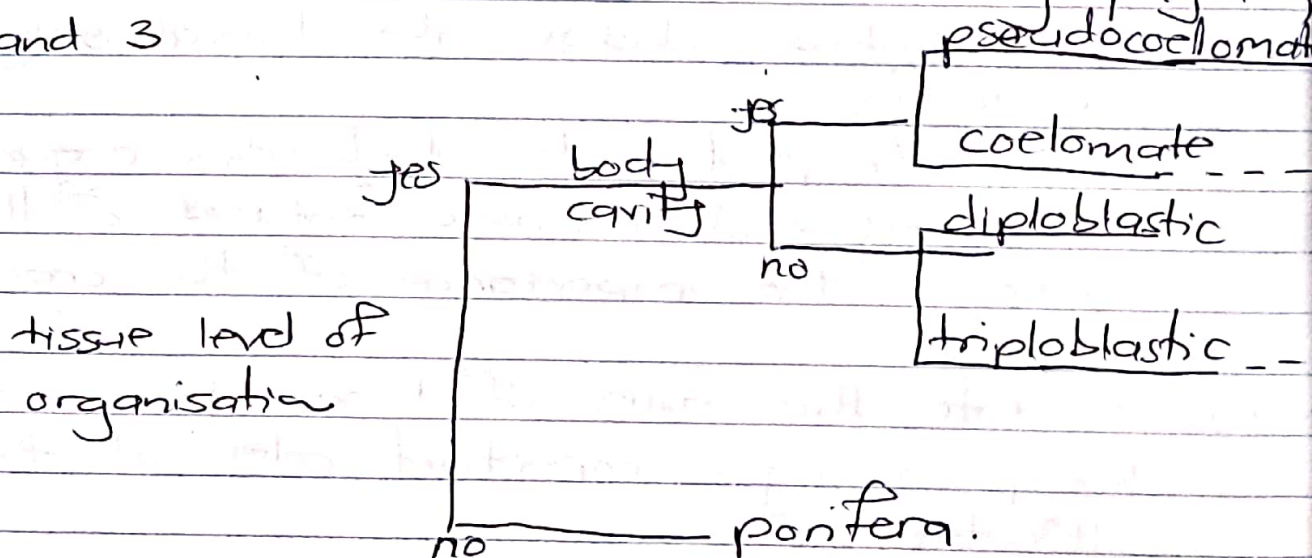


- Identify the phylum and genus of the organism [2]
- Describe the importance of the organisms, the phylum named in (a) [3]
- State any one diagnostic feature of organisms in the Kingdom Fungi [1]

J19/2 99. Complete table below by placing the organisms liver fluke, hookworm and leech into the phyla

Phylum	Organism
annelida	—
platyhelminthes	—
nematoda	—

- Fig below represents a dichotomous key. Use information in table above to identify phyla 1, 2 and 3



- 3/3 6c. Discuss the socio-economic importance of organisms in phylum Filicinophyta (ferns) [6]
- 2/3 6a. Outline the economic importance of kingdom Prokaryotae [8]
- 21/3 7a. Describe the importance of kingdom Plantae [8]
- 20/3 7a. Describe the economic importance of the kingdom Fungi [8]
- 20/3 6a. Outline the diagnostic features of phylum
1. Chordata in the animal kingdom
2. Angiospermophyta in the plant kingdom [8]
- 9/3 7a. Describe the general characteristics of the kingdom Protista [8]
b. Discuss the socio-economic importance of the kingdom Protista [6]
- 9/3 6c. Discuss steps to be taken when classifying an organism [6]
7c. Discuss the socio-economic importance of the kingdom Fungi
- 8/3 7a. Describe the characteristics of the kingdom Fungi [8]
b. Discuss the socio-economic importance of kingdom animalia [6]

- N23/1 40. The phylogenetic tree of animals show that
- A cnideria evolved from porifera
 - B flat worms evolved from roundworms
 - C both porifera and cnideria evolved from protoch
 - D both molluscs and annelids are segmented

- N22/1 37. Which is not a socio-economic importance the kingdom animalia?
- A tourism
 - B food
 - C fermentation
 - D hunting

38. Which is a diagnostic feature of the phylum Echinodermata?
- A triploblastic
 - B no head
 - C tube feet
 - D calcareous exoskeleton

39. Which one is the correct taxonomic hierarchy?
- A kingdom, phylum, class, order, family, genus, species
 - B kingdom, class, phylum, order, family, genus, species
 - C kingdom, phylum, order, family, class, genus, species
 - D kingdom, phylum, class, family, order, genus, species

40. Which is not an importance of prokaryotes?
- A Bio-technology
 - B decomposition
 - C Fermentation
 - D penicillin production

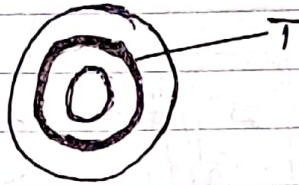
- N21/1 39. Which one is not a use of organisms found in kingdom Prokaryote?
- A Fermentation
 - B biotechnology
 - C decomposition
 - D tourism

40. Which importance is shared by prokaryotes and Fungi?
- A food
 - B clothing
 - C tourism
 - D spoilage

20/1 38 Which kingdom does not contain photosynthetic organisms.

- A. Fungi
- B. plantae
- C. protocista
- D. prokaryote

39. The diagram is a transverse section of the body of a cnidarian



What is tissue T

- A endoderm
- B ectoderm
- C mesoderm
- D mesoglea

40. Which one is a diagnostic feature of angiosperm

- A seeds are enclosed in an ovary
- B seeds are not enclosed in an ovary
- C sporophyte is the dominant generation
- D gametophyte is the dominant generation

20/1 30 Which one is a diagnostic feature of phylum coniferophyta?

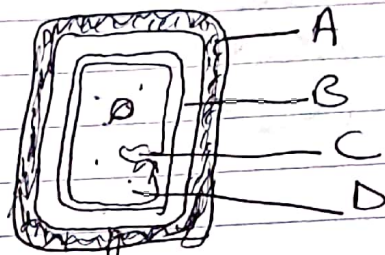
- A sporophyte generation is dominant
- B complex vascular tissue
- C seed lies on ovuliferous scales
- D no fruit because no ovary

40 Which phylum stores carbohydrates as mannitol.

- A. Sporozoa
- B. Oomycota
- C. Chlorophyta
- D. Phaeophyta.

N19/1 35. Two organisms which belong to different genera also belong to different
 A orders B species C families D classes

36. The diagram shows a bacterial cell. Which part A, B, C or D is the site of aerobic respiration?

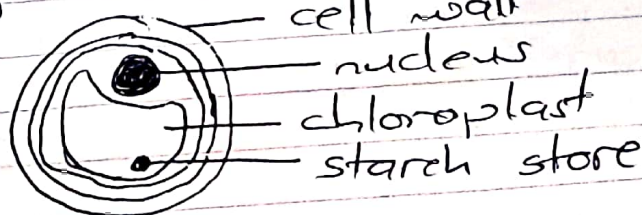


38. Which taxon in the list is the largest?
 A genus B family C order D species

40. Which kingdom is the major promoter of tax
 A. animalia B fungi C protista D prokaryotae

N2018/1 35. Which statement correctly describes alternation of generation in plants
 A. plants undergo evolution
 B. each type of plant has two life cycles
 C. two forms of the plant are present in the life cycle
 D. ~~increased~~ ^{plants} ~~carbon~~ undergo both sexual and asexual reproduction

36. The diagram shows a unicellular organism.



A. Prokaryote
C. Plantae

B. Protista
D. Fungi

38. Humans and apes belong to the same
A. genus B. species C. family D. order

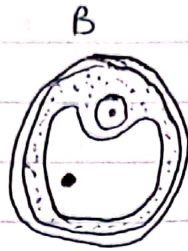
35. Which one is a diagnostic feature of Mollusca?
A. bilateral symmetry B. triploblastic
C. ventral muscular foot D. segmented body

36. Bryophytes can be regarded as the amphibians of the plant world because they
A. inhabit both terrestrial and aquatic habitats
B. live on land but have features associated with an aquatic habitat
C. rely entirely on water for growth and reproduction
D. live on land but need water for their reproduction

39. The socio-economic importance of kingdom Prokaryote is
A. medical use B. food spoilage
C. carbon sink D. leather tanning

40. Organisms of 2 different species cannot permanently occupy the same ecological niche because the 2 species
A. do not interbreed
B. may be part of separate food webs
C. would compete for the same resources
D. would have different nutritional requirements.

N24/1 38 Which organism belongs to the kingdom Fungi?



J25/1 34. Which feature is common to phylum Filicinophyta and phylum Basidiomycota?

- A. Carbohydrates stored as starch
- B. Cellulose cell wall
- C. Heterotrophic nutrition
- D. Reproduction by means of spores.

35. What is the socio economic importance of fungi that distinguishes it from prokaryotes?

- A. alcohol production
- B. decomposition of DOM
- C. food production
- D. experimental use.

36. Which of the following is not a biological significance of the coelom?

- A. acts as a hydrostatic skeleton in annelids
- B. provides a body cavity for growth and development of organs
- C. different parts of the gut can become differentiated for different functions.
- D. allows flattening of the body to overcome transport problems

37. The diagram shows an organism in the animal kingdom.



To which phylum does the organism belong?

- A. Annelida
- B. Arthropoda
- C. Cnidaria
- D. Platyhelminthes

24/2 9. Classification involves placing living organisms in a series of increasingly specialised groups based on similarities and differences.

a. State the name given to identify the science of classification (1)

b. Name any one branch of classification (1)

c. Explain why scientists prefer using phenetic classification to phylogenetic classification (3)

24/3 20. Distinguish between protozoa and algae (6)

25/2 10. Place order, phylum, genus and family in taxonomic hierarchy starting with the smallest (1)

b. Explain the binomial nomenclature in classification of organisms (3)

c. Suggest the importance of the kingdom Plantae to ecosystems (2)

25/3 6. Discuss the economic importance of kingdom Fungi (6)

24/3 6. Discuss the adaptations of angiosperms to life on land (6)

25/3 6. Describe any four taxa, apart from the kingdom in the classification of living organisms (8)

b. Explain the binomial system of naming organisms in taxonomy (6)

c. Discuss the socio-economic importance of kingdom Plantae

25/1 3. Which one is a diagnostic feature of organisms in the phylum Cnidaria?

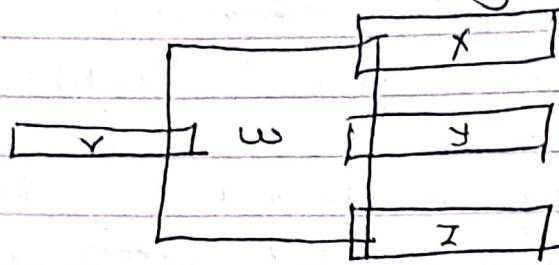
A. They are triploblastic

B. possess radial symmetry

C. exist as polyps and medusae

D. exhibit polymorphism.

35. The diagram shows one way of classifying organisms into the five kingdoms.



Which kingdom is represented by the box labelled w?

A. Fungi B. plantae C. prokaryotae D. protista

36. Which taxon has the lowest diversity of living organisms?

A. genus B. phylum C. order D. family